

WHERE DO BITS MATTER IN WORLD MODELS? A TINY M4 STUDY OF ENCODER-PRESERVING QUANTIZATION FOR DINO-WM PLANNING

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ABSTRACT

Efficient spatial reasoning requires world models that remain reliable under strict memory and compute budgets. We study this problem on DINO-WM planning in Wall with a Mac-only, workshop-scale protocol, sweeping ten weight-only low-bit variants across uniform and mixed precision designs. Our central hypothesis is that near the low-bit failure frontier, preserving encoder precision is more important than uniformly reducing all modules. The results show a clear regime pattern: 8-bit variants match FP16 success (0.50), 3-bit variants collapse (0.00), and 4-bit is the transition regime where precision allocation matters. At 4-bit, encoder-preserving mixed quantization (encoder FP16, predictor INT4) reaches 0.167 success versus 0.083 for uniform INT4 under the same planning budget. These preliminary findings motivate a broader research direction on module-aware precision allocation for robust and efficient world-model planning.

1 INTRODUCTION

World-model planning is attractive for efficient spatial reasoning because control is optimized in latent space rather than directly in pixels. Recent systems show strong planning behavior when representation quality is preserved (Zhou et al., 2024; Oquab et al., 2023). At the same time, low-bit quantization is now a default deployment tool, and broad surveys report that many models remain robust at moderate precision before degrading sharply in ultra-low-bit regimes (Liu et al., 2025). Recent world-model-specific evidence further suggests that quantization sensitivity differs across modules (Fu et al., 2026).

This motivates a concrete workshop question: for world-model planning, is the key decision merely *how many bits*, or also *where those bits are allocated*? We focus on the encoder/predictor split in DINO-WM and test the hypothesis that representation modules should keep higher precision near the low-bit frontier. The study is intentionally small and reproducible on consumer hardware (M4 Mac), designed to provide a clear empirical signal and a publishable starting point for larger follow-up work.

Contributions.

- We present a ten-variant bit-width sweep (FP16, uniform 8/4/3, mixed 8/4/3, plus E8/P4, E8/P3, E4/P3) for DINO-WM planning on Wall.
- We identify a three-regime pattern: 8-bit stable, 3-bit collapsed, and 4-bit as a transition regime.
- We show that at 4-bit, encoder-preserving mixed quantization doubles success over uniform INT4 under the same planning budget (0.167 vs 0.083), yielding a concrete hypothesis for future module-aware quantization research.

2 SETUP

We evaluate pretrained DINO-WM (Zhou et al., 2024) on the Wall environment using the repository’s native success metric from planning outputs. We do not redefine task scoring. The study

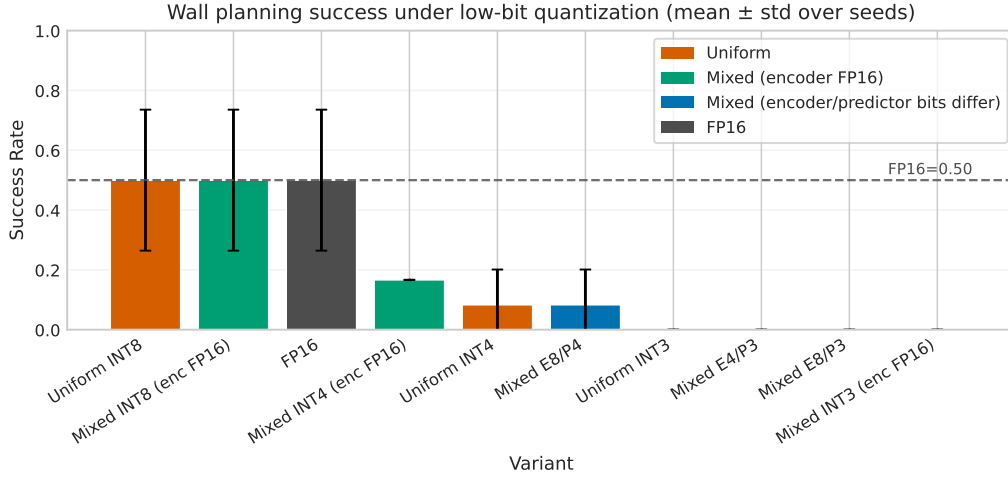


Figure 1: Success under low-bit quantization on Wall (mean $\pm$ std over two seeds). The transition appears at 4-bit: mixed INT4 (encoder FP16, predictor INT4) outperforms uniform INT4, while 3-bit variants collapse.

is intentionally compute-bounded: one environment, one checkpoint, one planner budget, and two seeds.

Variants. We compare ten variants. The baseline is FP16. Uniform variants quantize both encoder and predictor at 8, 4, or 3 bits. Encoder-preserving mixed variants keep the encoder at FP16 and quantize only the predictor (8, 4, or 3 bits). We also include three mixed encoder/predictor combinations: E8/P4, E8/P3, and E4/P3. All quantization is weight-only on linear layers.

Evaluation protocol. We run on Apple M4 (MPS backend) with `goal_source=random_state`, `goal_H=9`, `planner.opt_steps=2`, `planner.max_iter=2`, seeds $\{1, 2\}$, and $n_{evals} = 6$ episodes per seed. This yields 120 evaluated episodes total (10 variants $\times$ 2 seeds $\times$ 6 episodes). Per run we log success rate, average planning time, and model-size estimate under effective bitwidth accounting.

3 RESULTS

Figure 1 summarizes success rates. Three regimes emerge. First, 8-bit is stable: FP16, uniform INT8, and mixed INT8 all reach mean success 0.50. Second, 3-bit collapses: both uniform INT3 and mixed INT3 reach 0.00. Third, 4-bit is the transition region: mixed INT4 reaches 0.167 versus 0.083 for uniform INT4. This is a doubling in absolute success under the same planner budget.

The extended mixed settings refine this picture. Mixed E8/P4 reaches 0.083 (below encoder-preserving mixed INT4), while E8/P3 and E4/P3 are both 0.00. These comparisons indicate that simply assigning asymmetric bits is insufficient; keeping the encoder at higher precision appears important near the low-bit frontier.

Figure 2 shows the accuracy-memory frontier. Low-bit models reduce size substantially, but the smallest models (3-bit) fail on this task. Among 4-bit-class settings, encoder-preserving mixed INT4 provides the best success in our runs, albeit at larger size than uniform INT4. Table 1 reports key numbers.

4 DEMO AND REPRODUCIBILITY

We provide a replay-based viewer that loads saved trajectories and toggles variants without requiring GPU inference. The experiment pipeline is scriptable end-to-end: variant definitions live in

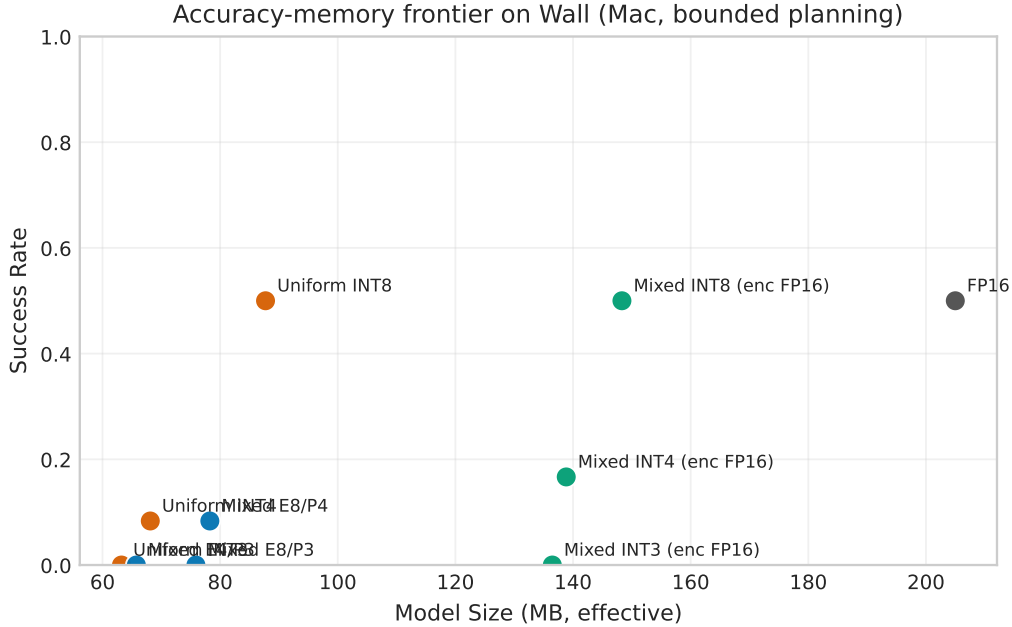


Figure 2: Accuracy-memory frontier for all ten variants. 8-bit variants retain FP16-level success in this setup; 3-bit variants fail; 4-bit variants differentiate by precision allocation.

Variant	Success	Time (s)	Size (MB)
FP16	0.500	54.15	204.99
Uniform INT8	0.500	55.49	87.72
Mixed INT8 (enc FP16)	0.500	54.46	148.31
Uniform INT4	0.083	54.99	68.12
Mixed INT4 (enc FP16)	0.167	52.74	138.84
Uniform INT3	0.000	52.52	63.23
Mixed INT3 (enc FP16)	0.000	53.21	136.47

Table 1: Key results at opt_steps=2, goal_H=9 (means over two seeds).

one config file, each run writes a consistent metrics JSON, and aggregation/figure scripts produce publication figures directly.

5 LIMITATIONS AND OUTLOOK

This is a preliminary tiny-paper study and should be interpreted as such. The sample size is small (two seeds), confidence intervals are wide at 8-bit, and the conclusions are limited to Wall with one planner budget. Our quantization path is weight-only and does not include activation quantization or hardware-fused low-bit kernels. As a result, the timing differences are modest in this Mac-only setup and should be interpreted primarily as a controlled robustness/compression study. Nevertheless, the threshold behavior and 4-bit encoder-sensitivity pattern provide a concrete next-step hypothesis for broader spatial reasoning benchmarks: as bitwidth approaches the failure frontier, precision should be allocated preferentially to representation modules rather than uniformly across the world model.

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